

# ERNEST WINDEL DREO

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## SUMMARY

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Fresh Computer Science graduate with hands-on experience through software engineering and technical internships at Zetheta Algorithms Private Limited and the Department of Information and Communications Technology (DICT). Skilled in Node.js, TypeScript, Python, PostgreSQL, Redis, Kafka, API development, software testing, troubleshooting, and system support. Experienced in system design and architecture, distributed systems, transaction processing, data pipelines, reconciliation, and database systems, with a strong foundation in designing and building reliable, scalable, and maintainable applications.

## EDUCATION

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### SORSOGON STATE UNIVERSITY – BULAN

Sorsogon City  
2026

BS Computer Science

Major in Software Engineering · Minor in Data Science

Relevant Coursework: Data Structures & Algorithms, Software Engineering, Operating Systems, Database Systems, Web & Mobile Development

## INTERNSHIP EXPERIENCE

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### ZETHETA ALGORITHMS PRIVATE LIMITED

Remote

Software Engineer

July 2026 – Aug 2026

- Designed a distributed fintech transaction-processing architecture using Kafka, PostgreSQL, and Redis, targeting 12,000+ transactions per second (TPS) through event streaming, database sharding, and caching.
- Developed a Node.js/TypeScript transaction service with ACID-compliant transfers, idempotency, and concurrency controls, improving transaction reliability and preventing duplicate processing.
- Architected Kafka-based account partitioning and saga patterns for cross-shard transactions, supporting scalable transaction processing without distributed locks on the primary processing path.
- Built a Python data-ingestion pipeline for SWIFT MT940, ISO 20022 CAMT.053 XML, and bank-specific CSV files, standardizing multiple banking formats into a unified transaction schema.
- Documented system architecture and transaction workflows using draw.io, clearly illustrating service interactions, data flows, and infrastructure components for technical review and implementation.

### DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

Sorsogon City

Information Technology (IT) Intern – DICT Region V

July 2025 – Aug 2025

- Assisted in Computer assembly, hardware troubleshooting, Windows Installation, and system maintenance.
- Configure wired and wireless network connections, including RJ45 cable setup and Starlink internet installation.
- Supported digital literacy training and eGov Super App onboarding Activities.
- Prepared technical documentation, reports, and presentation materials.
- Coordinated virtual meetings with Local Government Units using Zoom.
- Assisted in organizing DICT technology programs and ICT events.

## PROJECTS

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### Learner's AI – Course Recommendation with Explainable AI (Thesis Project)

2026

Lead Developer

- Engineered an AI-driven course recommendation platform that personalizes learning pathways using learner interests, skills, and academic preferences.
- Develop a React.js frontend and FastAPI/Node.js backend architecture for recommendation delivery, AI inference, and API integration.
- Integrated Large Language Models (LLMs) to provide natural-language explanations for individual course recommendations.
- Implemented Explainable AI (XAI) mechanisms to make recommendation logic more transparent and interpretable to learners.
- Designed the system to combine personalized recommendations with explanations, enabling users to understand the reasoning behind suggested courses.

Lead developer

- Develop an AI-powered text classification system for detecting fake news and spam content using Natural Language Processing and machine learning techniques.
- Implemented a 5-stage text preprocessing pipeline with tokenization, lowercasing, stopword removal, punctuation handling, and feature extraction.
- Engineered 11+ text-mining features, combining TF-IDF patterns with linguistic and sentiment-based indicators for classification.
- Built and evaluated a Decision Tree classifier, achieving 87.5% accuracy and 87.1% F1-score on the project's test dataset.
- Developed an interactive web interface with HTML, CSS, JavaScript, and Chart.js for real-time text analysis, feature-importance visualization, decision-tree interpretation, and model evaluation.
- Added dataset management functionality allowing users to inspect training data and upload custom CSV datasets for model retraining.

## LEADERSHIP EXPERIENCE

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### SUPREME STUDENT COUNCIL

Bulan, Sorsogon

BS Computer Science Representative

2024 – 2025

- Represented Computer Science students in university activities.
- Participated in planning and coordinating student initiatives.

### COMPUTING SOCIETY (ComSoc)

Bulan, Sorsogon

Media Information Officer

2023 – 2024

- Created promotional materials and announcements for organization activities.

## ADDITIONAL

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**Technical Skills:** JavaScript, TypeScript, Python, SQL, React, Next.js, Node.js, Express.js, FastAPI, REST API Development, HTML5, CSS3, Tailwind CSS, PostgreSQL, MySQL, MongoDB, Redis, Kafka, Git, GitHub, Postman, Supabase, Railway, Vercel, Machine Learning, Large Language Models (LLMs)

**AI & Developer Tools:** ChatGPT, Claude Code, OpenAI Codex, DeepSeek, OpenRouter, AI-Assisted Development, Prompt Engineering, LLM-Based Application Development

**Certifications:** Cisco Networking Academy – Computer Hardware Basics, Operating System Basics, Endpoint Security; IBM/Coursera – Developing Back-End Apps with Node.js and Express; Zetheta Algorithms Private Limited – Software Engineer HTTP; DICT Internship Program – Apprenticeship Certificate

**Additional Certificates:** Sololearn – Prompt Engineering, Introduction to LLMs, Web Development, JavaScript Intermediate, Introduction to JavaScript, Introduction to Python, Introduction to HTML, Introduction to CSS

Languages: English (B2), Filipino (native)